

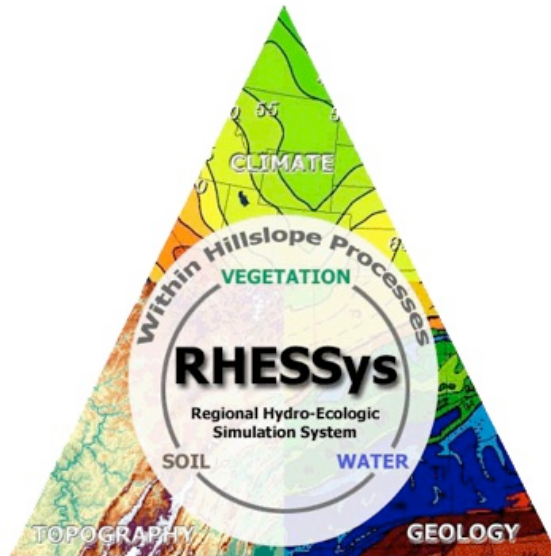
Welcome to the RHESSys Manual

(under development)

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Purpose

Include info about the manual, why it exists, what it is intended for

Scope

- What to expect from this book (user will learn....)
- What the manual covers (practical step-by-step guidance)
- What it does not cover (underlying theory of the model, all possible functionality)

Intended Audience

- Who this is for/Who should be reading this

Assumed basic prerequisite knowledge

- General Modeling
- Basic R/Rstudio

- **GIS and Raster/Vector**
 - **Basic Unix/Linux environment** (shell, command line, text editor)
 - **Git/GitHub**
-

title: “About this Manual”

Conventions

- How to read this document
- Key/Legend for document formatting - regarding use of icons, symbols, different font indications

Manual Structure

- organization
- navigation
- links to sections
- potentially a link to a quick start guide

Support/Resources

- Potentially include author info (Naomi, Janet (contact info?), Ryan)
- RHESSys GitHub repositories (RHESSys code, RHESSysPreProcessing, RHESSysIOinR, ParameterLibrary)
- RHESSys Wiki(s)
- Potentially a FAQ page, a user community forum (i.e. ‘issues’ section on RHESSys Github repo?)

title: “Part I: Foundations & Environment”

- This part is about understanding the logic of this tool and setting the computational environment
- intro to what is covered in the section
- expected outcome for this section: functioning modeling environment and clear strategic plan

title: “1 Introduction (The Model)”

1.1 Overview of RHESSys

What RHESSys is

- Description

Capabilities

- Integrated Ecohydrological Modeling
- Spatial distribution of the landscape
- Feedback loops
- Fire model

Limitations

- High data demands
- Can be computationally intensive
- Maybe any processes that are not well represented? Let users know if the model won't fit their research question

1.2 Recommended reading/Publications

- Foundational papers
- Overview/Review papers
- Applications

1.3 Workflow overview

- Step 1: Installation - Set up the RHESSys environment
- Step 2: Data Preparation - Gather and format spatial, climate, and validation data

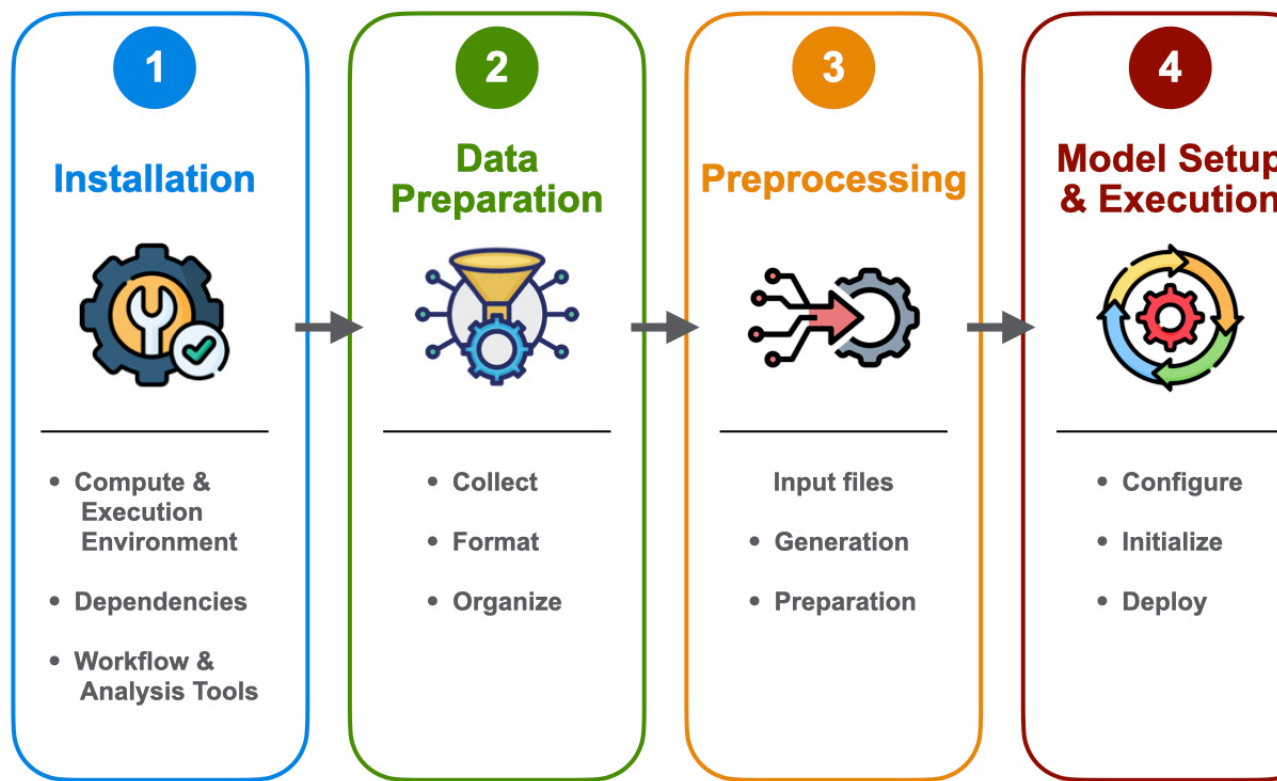


Figure 1: RHESys Modeling Workflow Overview

- Step 3: Preprocessing - Generate the landscape representation (Worldfile) and routing (Flowtable) input files
- Step 4: Model Setup and Execution - Define the simulation rules and run RHESSys
- Step 5: Calibration & Sensitivity Analysis - Condition the model (Initialize/reach equilibrium), refine parameters, validate performance
- Step 6: Post Processing - wrangle, analyze, and visualize output data

Key concepts and terminology

(define frequently used terms)

- Spatial hierarchy (basin, hillslope, zone, patch, stratum)
- Input File types (Worldfile, Header, Flowtable, Tecfile, Parameter/Climate files, Output filters)
- State variables (store quantities) and fluxes
- Spatial/Temporal step (i.e., basin daily, patch monthly)

title: “2 Setup (The Software)”

2.1 Setup options

(description of each, who should use, when to use, recommendation, link to corresponding install instructions)

2.1.1 Local Installation (MacOS, Linux)

- more advanced users
- users who want more flexibility

2.1.2 Containerized setup (Docker)

- beginners
- Windows users
- portability

2.1.3 HPC/Cloud (SIF/Singularity)

- high-performance computing environments
- cluster users
- large scale, high resolution simulations

2.2 RHESSys Installation

2.2.1 MacOS (include section for Linux as well if necessary or where differs)

Code Dependencies and Operational Requirements

- Terminal application
- Xcode Command Line Tools (compilers/build tools)

- Privileges (i.e. Administrator access)
- Homebrew
- Git
- Libraries (i.e. NetCDF)

Installation steps

- Download code from GitHub
 - Zip file
 - Clone repository (main, develop)
- Compile
- Shell configuration (adding to path)
- Verify installation
- Common issue and solutions

2.2.2 Docker

- Install Docker Desktop
- Obtaining RHESSys image from the Docker Hub

2.2.3 SIF

- Obtaining RHESSys .sif image

2.3 Installing Workflow & Analysis Tools

2.3.1 R Ecosystem

- R (The engine)
- RStudio (The interface)
- Standard Packages (Extensions - Libraries that expand functionality)

2.3.2 WhiteboxTools for Geospatial Processing

2.3.3 RHESSys-Specific R packages

- RHESSysPreprocessing
- RHESSysIOinR

